

PHYS 427: Lecture # 9

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Lecture Span

1. Thermodynamic potentials

Recap

For reversible processes, $dQ = TdS$.

For irreversible processes $dQ < TdS$

Thermodynamic Potentials

Up to this point, we have covered two thermodynamic potentials:

(Fixed temperature and volume)

$$F = U - TS, \quad U \quad (1)$$

For some intuition, F represents the maximal work you can extract from the system.

Let $U = Q + W$, then for irreversible processes, we obtain

$$W_{on} > \Delta F \quad (2)$$

Today, we will cover two more potentials. We define enthalpy:

(fixed pressure)

$$H = U + pV \quad (3)$$

The final energy is Gibbs Free Energy:

(temperature and pressure fixed)

$$G = U + pV - TS \quad (4)$$

These are the maximum amount of work you can take out of a system (reversible process). Then,

$$\Delta G = \Delta U + p\Delta V - T\Delta S \quad (5)$$

$$= Q + W + p\Delta V - T\Delta S \quad (6)$$

$$= W + p\Delta V \quad (7)$$

Then, ΔG would be the extra work you need to do on top of the ideal work in order to push work out of a system.

Now, we have four quantities in the form of thermodynamic potentials:

1. $U(S, V)$; $dU = TdS - pdV$ (thermodynamic identity)
2. Helmholtz free energy: $F(T, V) = U - TS$

$$dF = dU - TdS - SdT \iff dF = -pdV - SdT$$

3. Enthalpy: $H(p, S) = U + pV$, then

$$dH = dU + pdV + Vdp \iff dH = TdS + Vdp$$

4. Gibbs Free energy: $G(p, T) = U + pV - TS$, then

$$dG = Vdp - SdT$$

Maxwell's Relations

Use mixed partials equivalences to derive the relations. Note that,

$$T = \partial_S U, \quad p = -\partial_V U \quad (8)$$

Then,

$$\frac{\partial T}{\partial V} = -\frac{\partial p}{\partial S} \quad (9)$$

Also,

$$dF = -SdT - pdV \iff -S = \partial_T F, \quad -p = \partial_V F \quad (10)$$

Therefore,

$$\partial_V S = \partial_T p \quad (11)$$

Use dH to derive

$$\partial_p T = \partial_S V \quad (12)$$

Use dG to derive

$$\partial_T V = -\partial_p S \quad (13)$$

Example

Given gas, measure T, V . But we want S . Then, we know that the natural variables of S is U, V . But we want S in terms of T and V . We first note that:

$$dS = \partial_T S dT + \partial_V S dV \quad (14)$$

Then, introduce heat capacity.

$$C_V = \left(\frac{dQ}{dT} \right)_V = T \frac{\partial S}{\partial T} \quad (15)$$

Also,

$$\frac{\partial S}{\partial V} = \frac{\partial p}{\partial T} \quad (16)$$

Then,

$$dS = \partial_T S dT + \partial_V S dV \quad (17)$$

$$= \frac{C_V}{T} dT + (\partial_T p) dV \quad (18)$$

Use ideal gas law to obtain

$$dS = \frac{C_V}{T} dT + \frac{Nk_B}{V} dV \quad (19)$$

$$S_f - S_i = \frac{3}{2} Nk_B \ln(T_f/T_i) + Nk_B \ln(V_f/V_i) \quad (20)$$